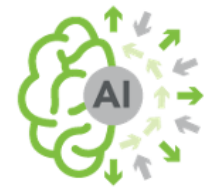
AI CLUSTERS



Cluster-less Validation of AI Accelerator Silicon Under Real Workloads

Matt Bergeron *Meta*

Ashutosh Kumar *Keysight Technologies*

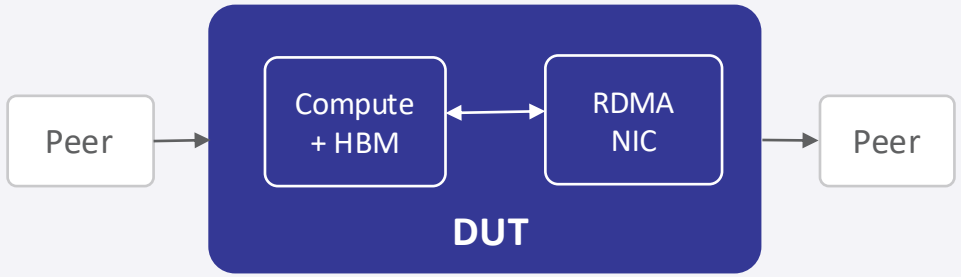
Cosmin-Nicolae Prunaru *Keysight Technologies*



The validation gap

Network-driven behavior dominates workload performance at scale — and is invisible in a bring-up lab.

Where validation happens

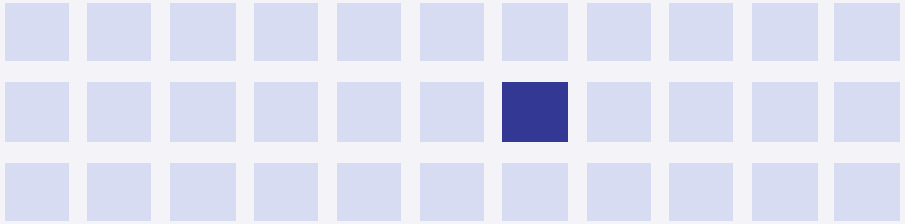


A few point-to-point peers: no real workloads, predictable congestion

gap

problems surfacing in cluster can only be fixed as a workaround

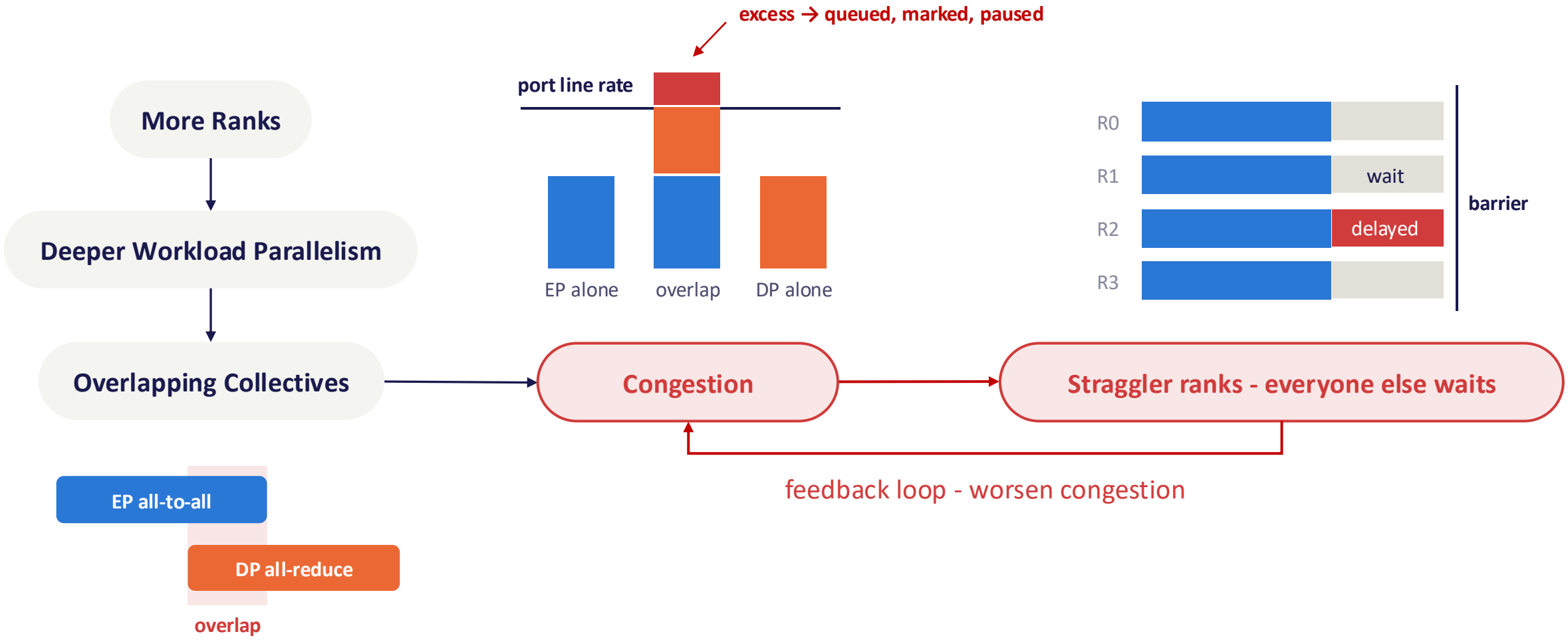
Where the chip runs (arrives much later)



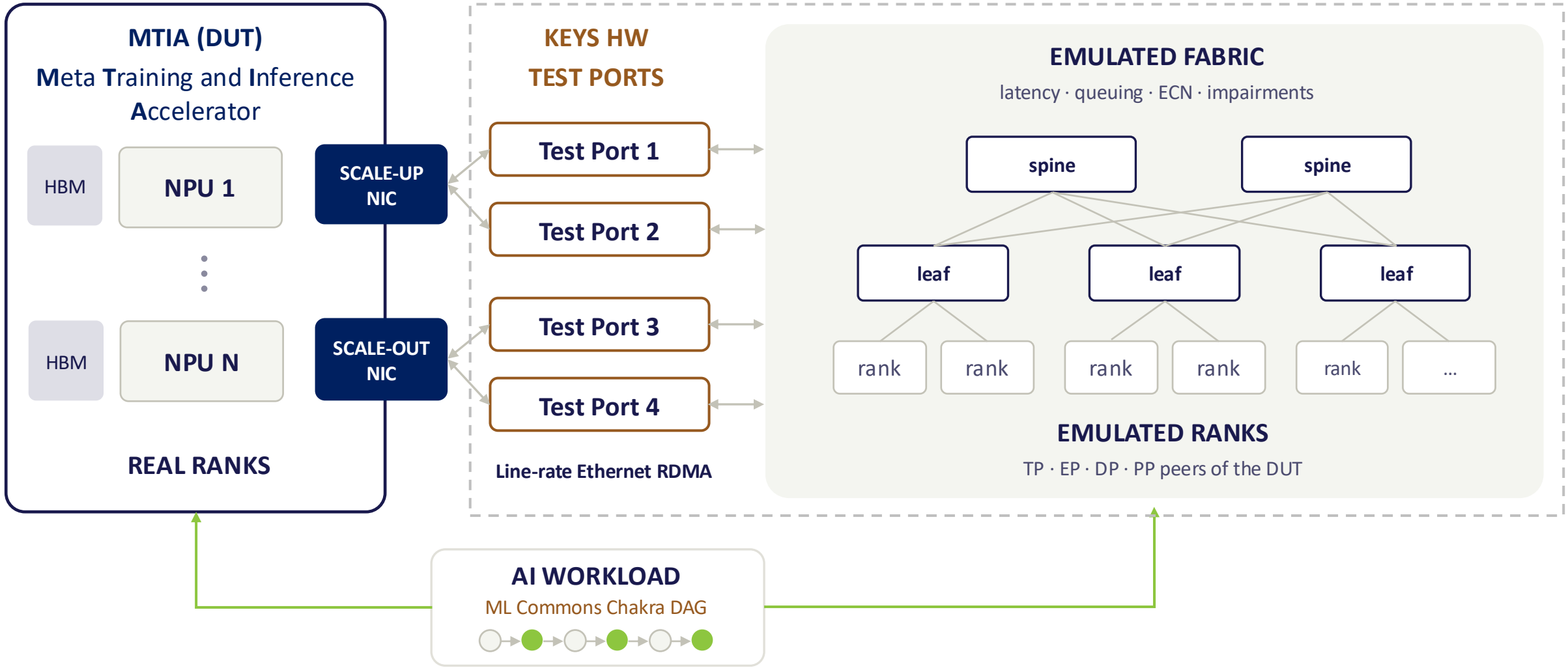
Thousands of peers: real workloads, unpredictable congestion

Architecture → **Pre-silicon** → **Tape-out** → **Bring-up lab** → **First cluster** → **Production**

What breaks at scale



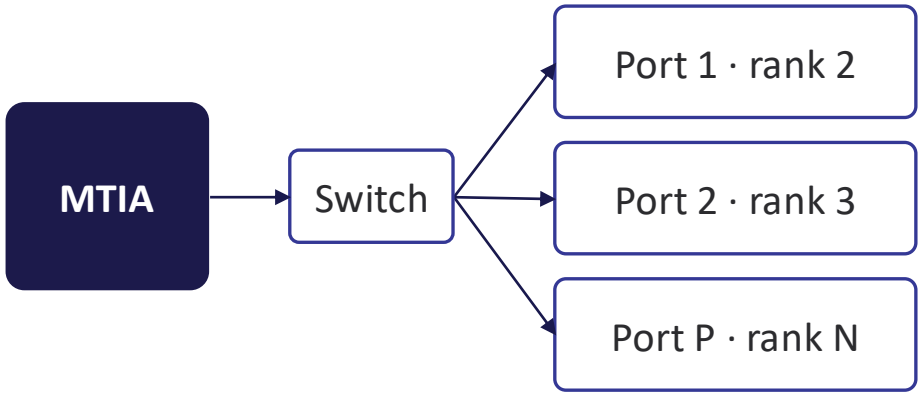
Recreating the fabric around MTIA



Two modes – bandwidth truth vs state truth

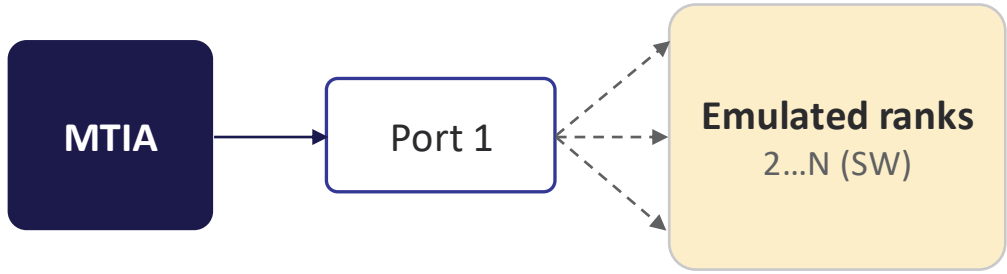
Test problems surfaced due to max bandwidth usage vs large scale of ranks

Emulated ranks — one per test port



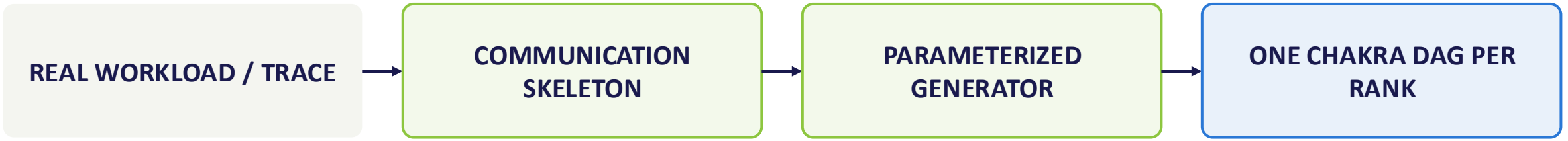
True per-rank bandwidth & convergence (in-cast); Scale ≤ port count

Emulated ranks — many behind one port



Exact per-rank connection or QP state at any scale; Port bandwidth shared

Repeatable patterns in AI workloads

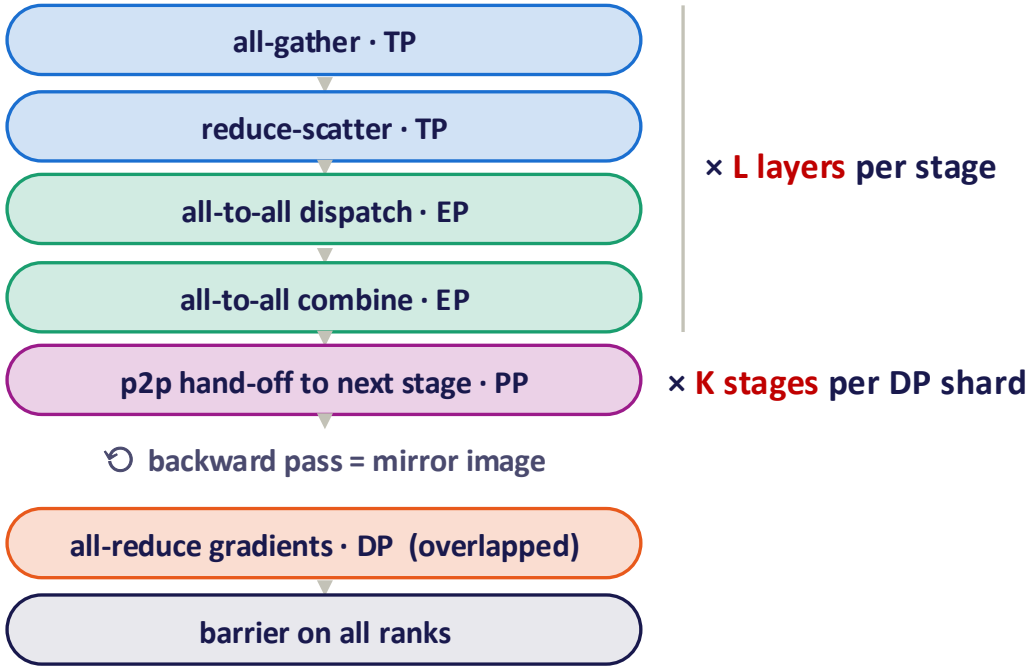


Training data

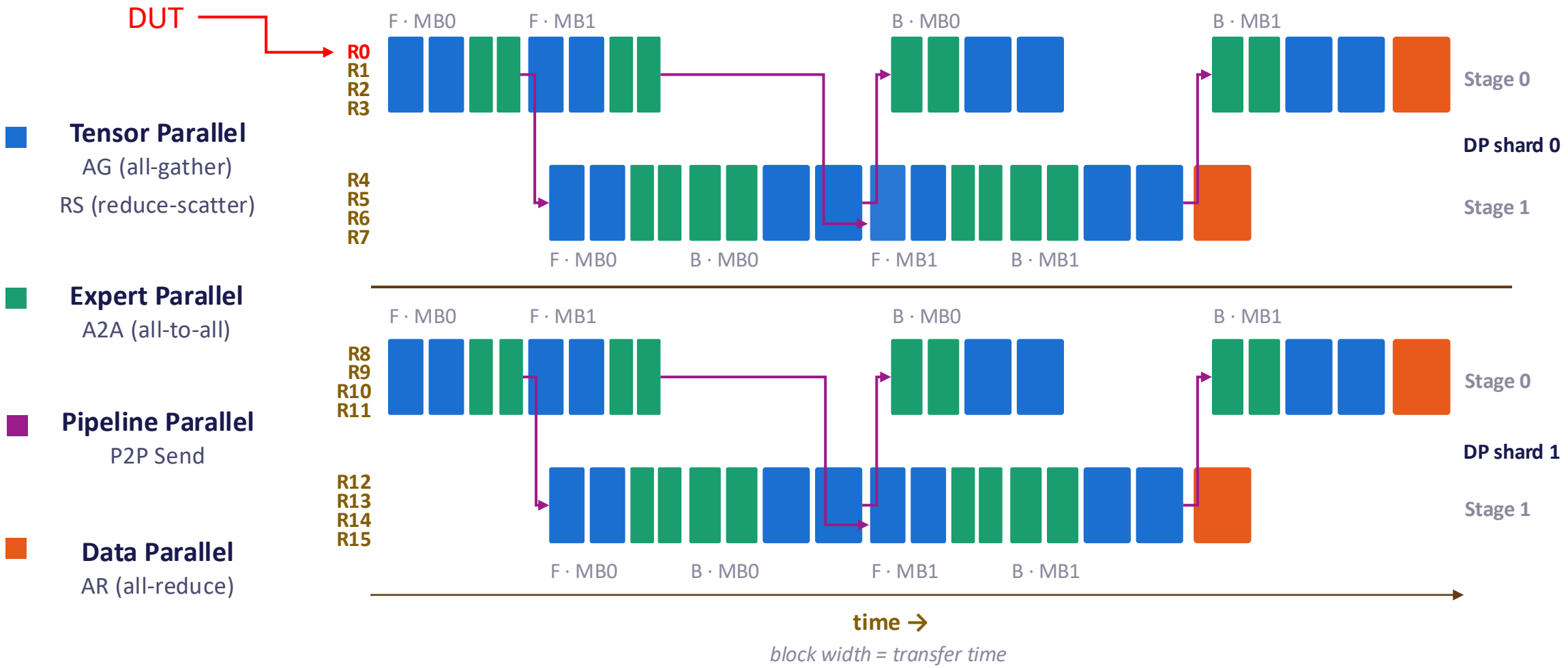
Sharded across racks

Split into M micro-batches per layer

N iterations of common repeatable sequence in training:



Communication sequence with 16 ranks



Execution with 32 ranks

Each rank exercises 1 scale-up NIC and 1 scale-out NIC exclusively

Run collectives in isolation

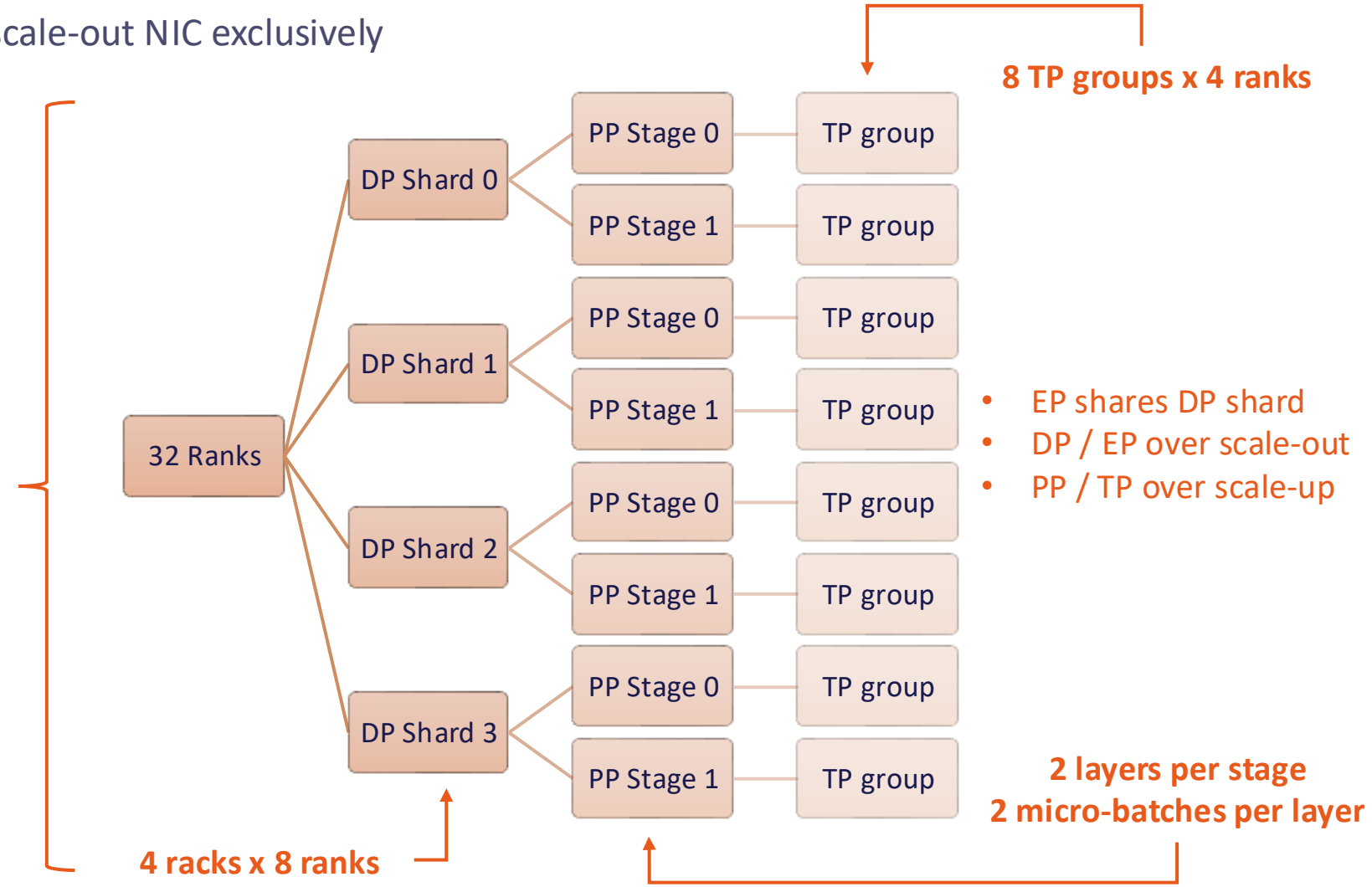
Run 100 iterations for each collective operation from the workload

Run training workload

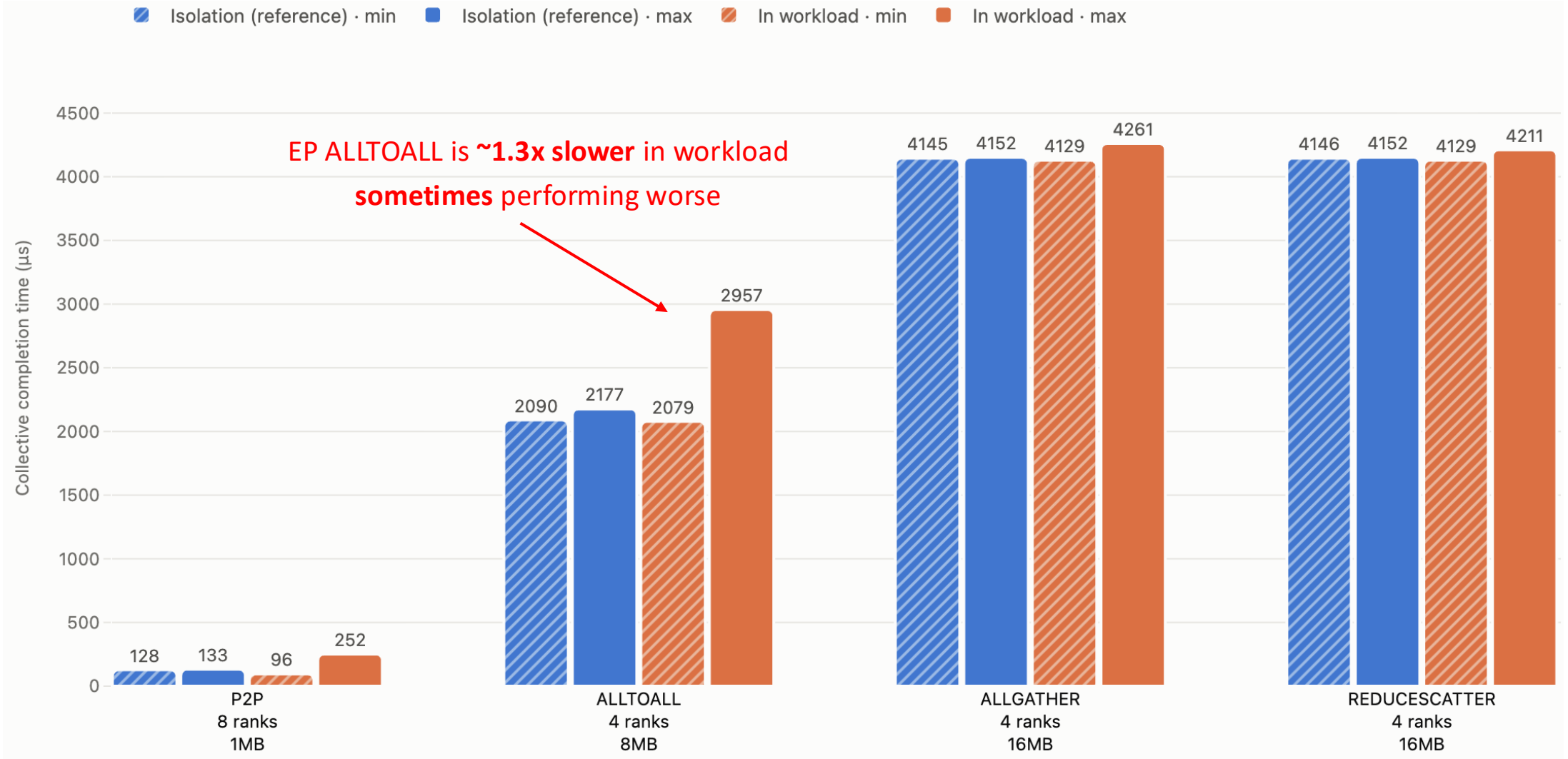
A repeatable pattern once consisting of mix of collective operations

Compare performance in isolation vs in workload

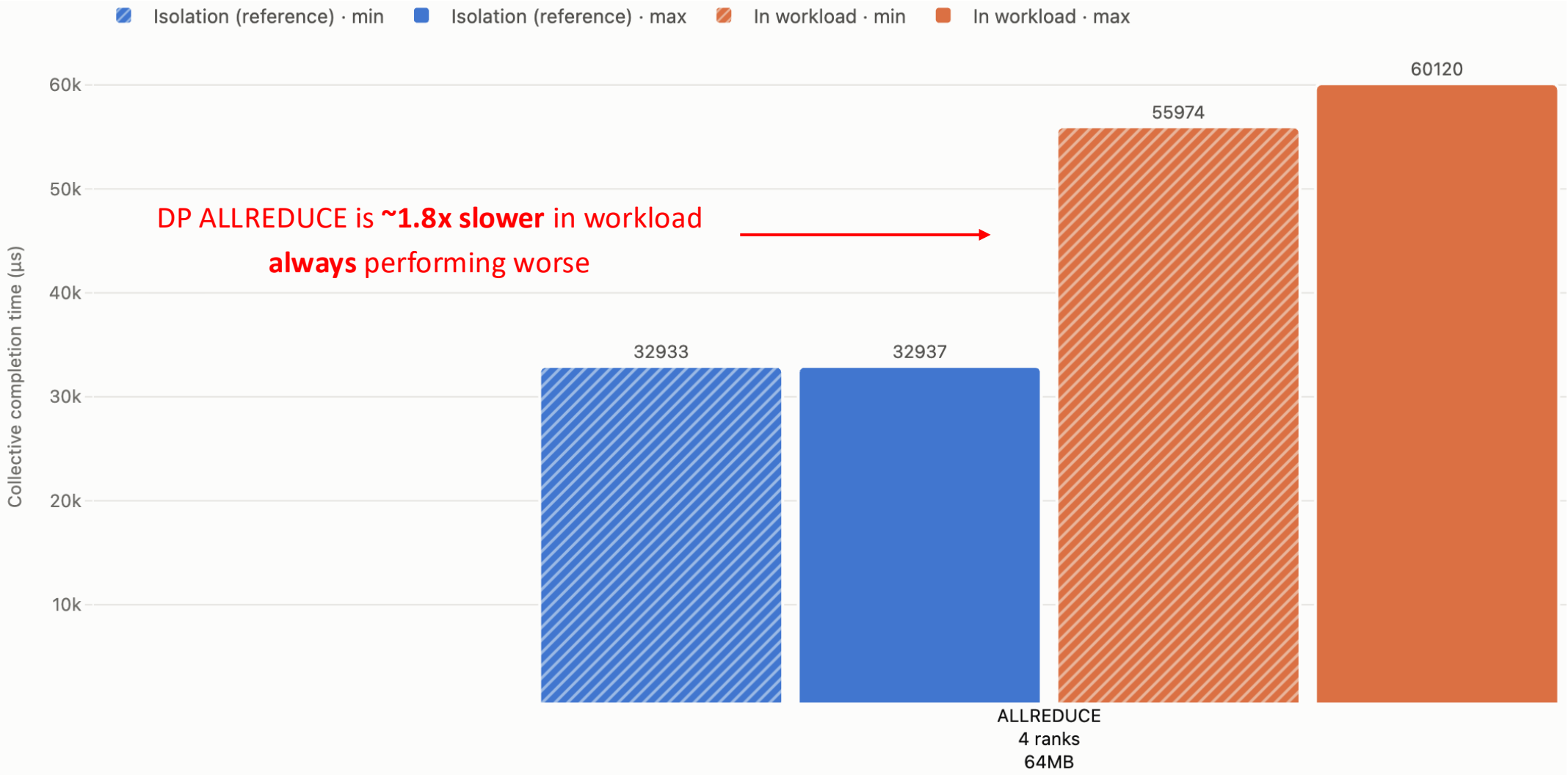
Group by – operation type, rank count, collective buffer size



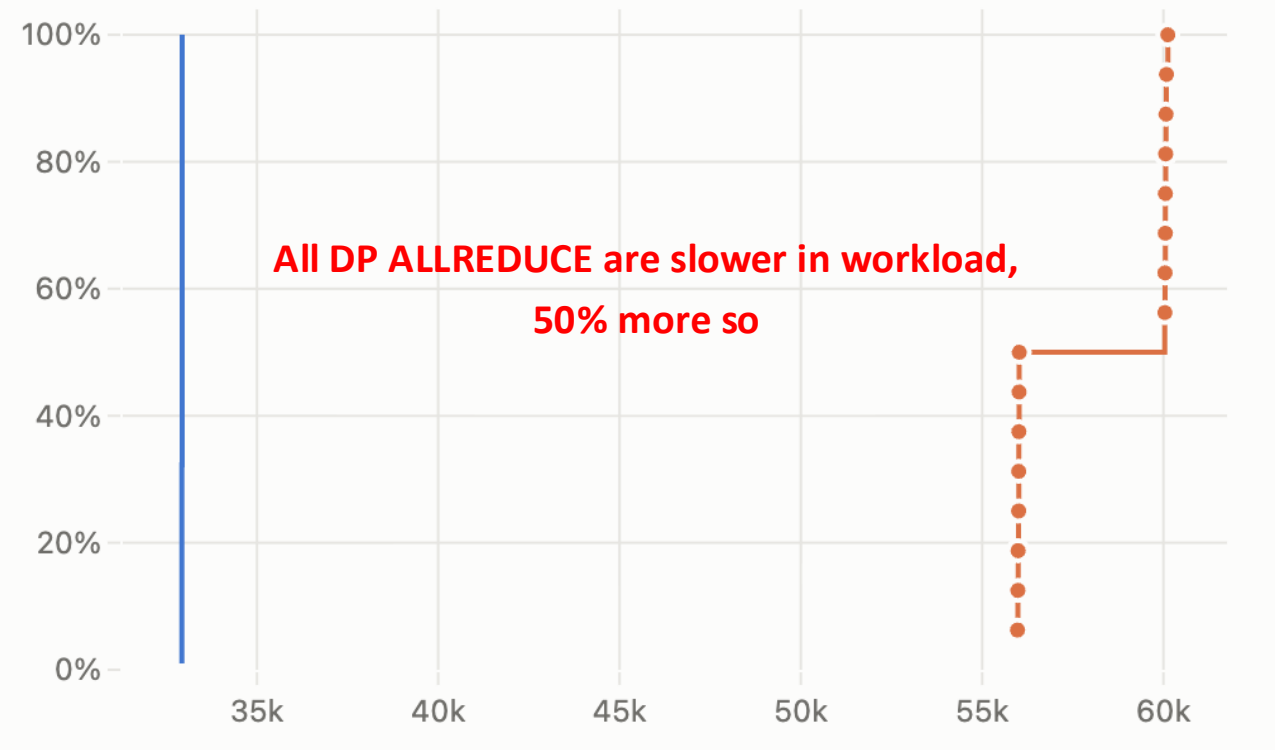
Compare performance for PP, EP, TP



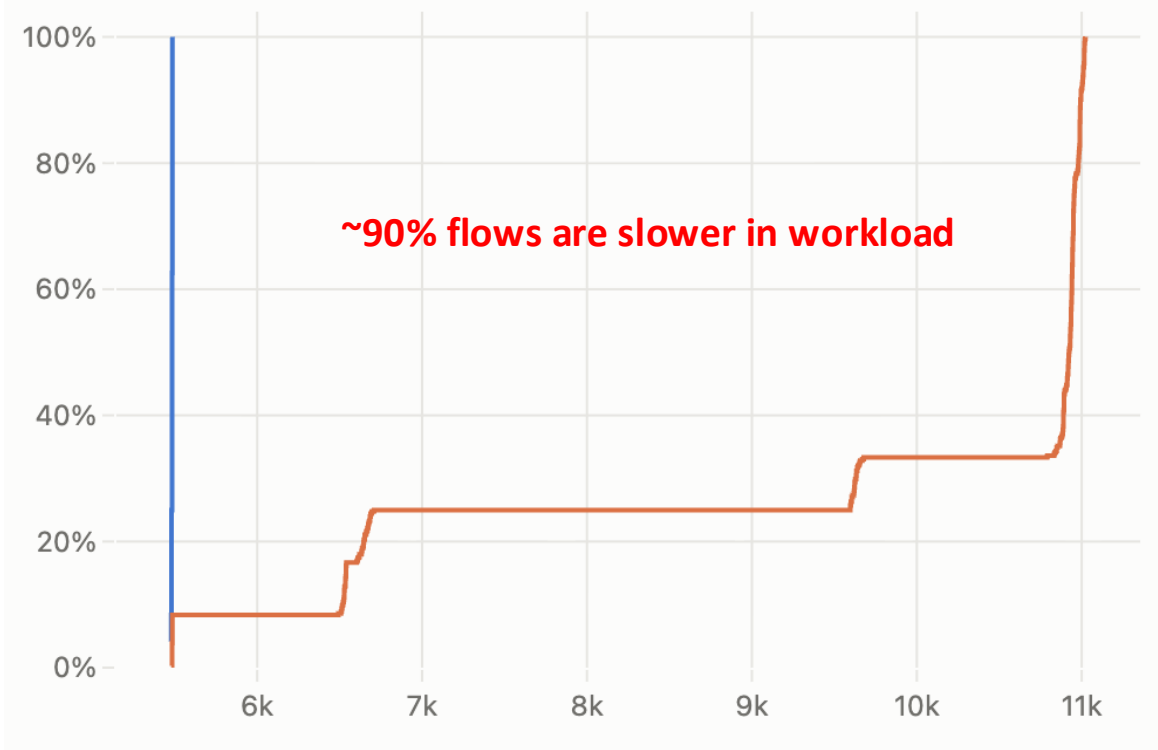
Compare performance for DP



Compare performance for DP - CDFs



Collective Completion Time

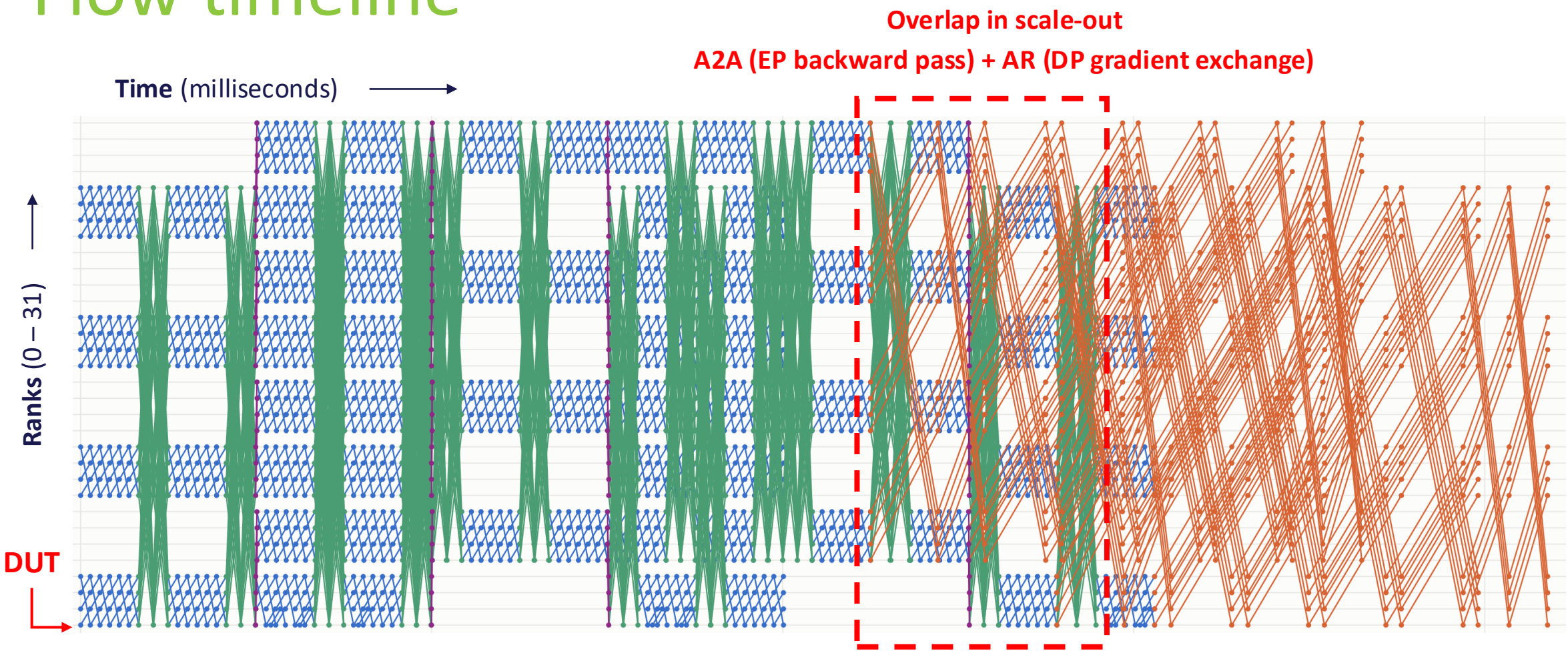


Flow Completion Time

Isolation (reference)

In workload

Flow timeline



Tensor Parallel
AG (all-gather) RS (reduce-scatter)

Expert Parallel
A2A (all-to-all)

Pipeline Parallel
P2P Send

Data Parallel
AR (all-reduce)

Call to Action

1. **Standardize tooling to generate DAGs capturing repeatable patterns for most workloads**
   Parameterized MVP DAG generation based on target topology and fabric

2. **Standardize the scorecard**
   MVP metrics and distributions to help root-cause issues faster

3. **More comprehensive fabric emulation**
   Multiple scale-up and scale-out NICs for each rank and impairment built around multi-tier latency, oversubscription, reordering

4. **Scale to 1K – 10K ranks**
   Execute the repeatable workload pattern on multiple ranks emulated behind single HW test port

Thank You!

For queries and demo, please reach out in person

Meta MTIA

Keysight AI